Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
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Project Report

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Project Title: Comparison of ML models for classification of heart disease

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Abstract

This project investigates the use of machine learning for heart disease prediction using the UCI Heart Disease dataset. Two dataset variants were compared: a Full dataset containing 920 records with fewer features and a Cleveland dataset containing 304 records while preserving all clinical variables. Data preprocessing included missing value handling, categorical encoding, scaling, and optional class balancing. Logistic Regression, Decision Tree, and Random Forest models were evaluated using stratified 5-fold cross-validation. The best performance was achieved by Logistic Regression on the unbalanced Cleveland dataset, obtaining an accuracy of 0.862, F1-score of 0.844, and ROC-AUC of 0.917. Important predictors included the number of major vessels observed by fluoroscopy, chest pain type, exercise-induced angina, and ST depression. The results show that preserving informative clinical features was more beneficial than increasing dataset size. Overall, simple interpretable models achieved strong predictive performance for heart disease classification.

1. Introduction

Cardiovascular disease (CVD) remains the leading cause of death worldwide. According to the World Health Organization, approximately 19.8 million people died from CVD in 2022, accounting for 32% of all global deaths. These statistics highlight the importance of early detection, which can enable timely intervention through treatment, lifestyle modification, and patient counselling before irreversible damage occurs. Recent advances in machine learning have demonstrated the potential for accurate heart disease prediction using routinely collected patient data. Such approaches may provide cost-effective decision support tools, particularly in regions with limited healthcare resources, by assisting clinicians in identifying high-risk patients at an early stage. This project investigates the effectiveness of machine learning algorithms and the influence of dataset characteristics on heart disease prediction using the UCI Heart Disease dataset available on Kaggle [3]. Two dataset variants were examined: a full dataset containing more patient records but fewer features, and a Cleveland-only dataset containing fewer records while retaining a richer set of clinical variables. Three classification algorithms—Logistic Regression, Decision Tree, and Random Forest—were evaluated and compared. Model performance was assessed using standard classification metrics, while stratified k-fold cross-validation was employed to provide robust and reliable evaluation.

This project aims to:

1. Develop and compare machine learning models for heart disease-prediction.
2. Evaluate the impact of dataset balancing on model performance.
3. Investigate whether feature-rich datasets perform better than larger datasets with fewer features.

References

- [1] World Health Organization. *Cardiovascular Diseases (CVDs) Fact Sheet, 2025. Accessed 01/06/2026.*
- [2] Md Manjurul Ahsan, Zahed Siddique, 2021, arXiv:2112.06459
- [3] UCI Heart Disease Data. Accessed 01/06/2026.

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2. Problem

The original dataset contains 14 features, obtained at 4 different locations, excluding origin of data and patient ID. Those columns were deemed not important for the study, as they are not representative of patient data.

Two separate datasets were created by cleaning and cropping the original dataset, to investigate if the number of features or number of records is more important in obtaining better results. The first one contains all the samples (later called full), from all the origins, however it has 11 features – we excluded slope, ca and thal, as those had many missing values, primarily in locations other than Cleveland. Second dataset contains only records, where origin is Cleveland (later called Cleveland) and preserves all the features but has approximately 60% less records. In full dataset 2 duplicates appeared due to feature collision after removing the abovementioned features. The duplicates were removed. Both dataset were tested with and without dataset balancing.

Originally, the dataset included multiclass classification of the stage of the CVD (0-4), however we have focused only on predicting the presence, not stage, of the disease. Therefore, we have modified the num variable to include only binary values with 1 signifying CVD is present, regardless of the stage. Table 1 summarizes the problem and dataset.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Binary classification
Dataset	Full – 920 samples, 10 features, Cleveland – 304 samples, 13 features
Target variable	Num – presence of heart disease (0 = healthy, 1 = CVD present)
Features	Original dataset contains 14 features: 6 numerical, 8 categorical — see Section 2.1
Missing values (%)	Original: Full: Cleveland: total: 11.95 5.20 0.21
Class distribution (%)	Full: Cleveland: 1 55.326087 45.723684 0 44.673913 54.276316
Evaluation metric(s)	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Evaluation strategy	Stratified 5-fold cross-validation

2.1 Exploratory Data Analysis

Each created dataset was explored separately. Class distribution, shown in Table 1, indicates only mild imbalance in both subsets. In the full dataset, patients with heart disease (class 1) are slightly more frequent, while in the Cleveland subset, healthy patients (class 0) form a small majority. Missing values, shown in Table 2, were also compared to the original dataset. Table omits features with no missing values. The table clearly shows that majority of missing data comes from locations different than Cleveland.

Table 2. Missing values

Feature	Original	Full dataset	Full (class 0)	Full (class 1)	Cleveland	Cleveland (class 0)	Cleveland (class 1)
total	11.95	5.20	2.96	7.00	0.21	0.30	0.10
trestbps	6.41	6.52	9.49	32.02	—	—	—
chol	3.26	21.96	9.49	32.02	—	—	—
fbs	9.78	9.78	3.41	14.93	—	—	—

restecg	0.22	0.22	0.00	0.39	—	—	—
thalch	5.98	5.98	4.87	6.88	—	—	—
exang	5.98	5.98	4.87	6.88	—	—	--
oldpeak	6.74	6.74	5.11	8.06	—	—	—
slope	33.59	—	—	—	0.33	0.61	0.00
ca	66.41	—	—	—	1.64	2.42	0.72
thal	52.83	—	—	—	0.99	1.21	0.72

The decrease in missing values is explained by removing part of the records. Increase is explained by data cleaning. In the original dataset part of the numerical feature records were filled with 0. We assumed these are denoting missing values, as it is impossible to have resting blood pressure or cholesterol of 0. During cleaning, the 0 values were changed into NaN giving a more accurate estimate of missing value numbers.

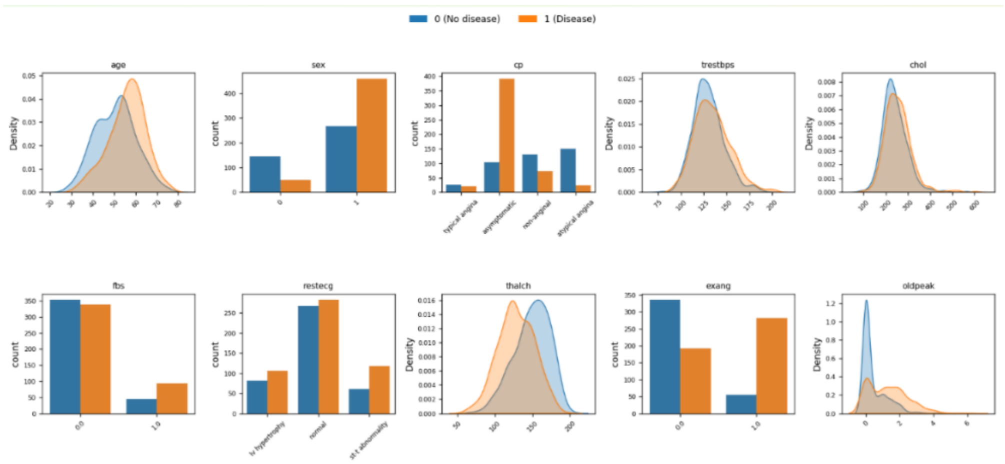

Figure 1. Feature distribution full subset

Figures 1 and 2 show the feature distributions for each subset. Overall, most features exhibit similar distributional shapes across the Cleveland and full datasets, suggesting general consistency in the underlying data generation process. The most notable differences are observed in the ST segment abnormality feature, which is rarely present in the Cleveland subset, and in thalch, where class 1 exhibits a more left-skewed distribution in the Cleveland subset.

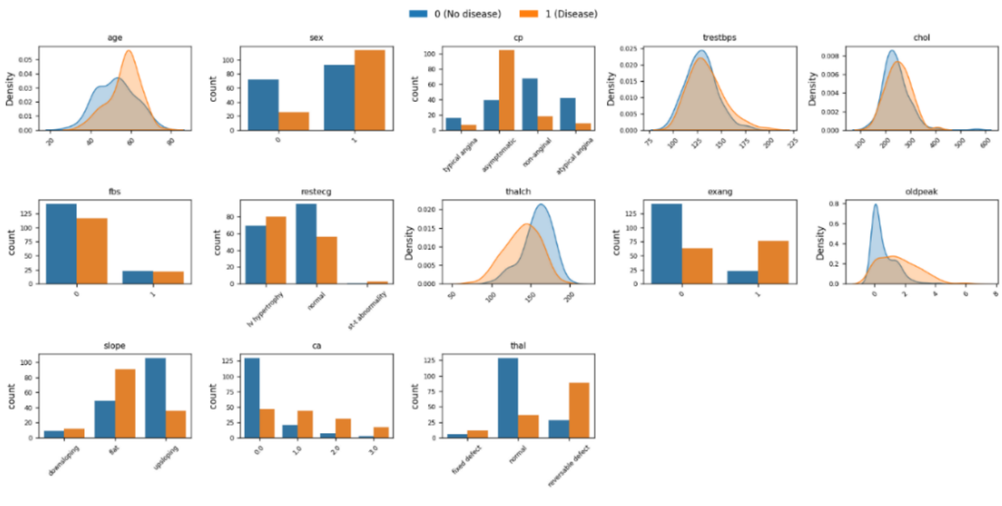

Figure 2. Feature distribution Cleveland subset

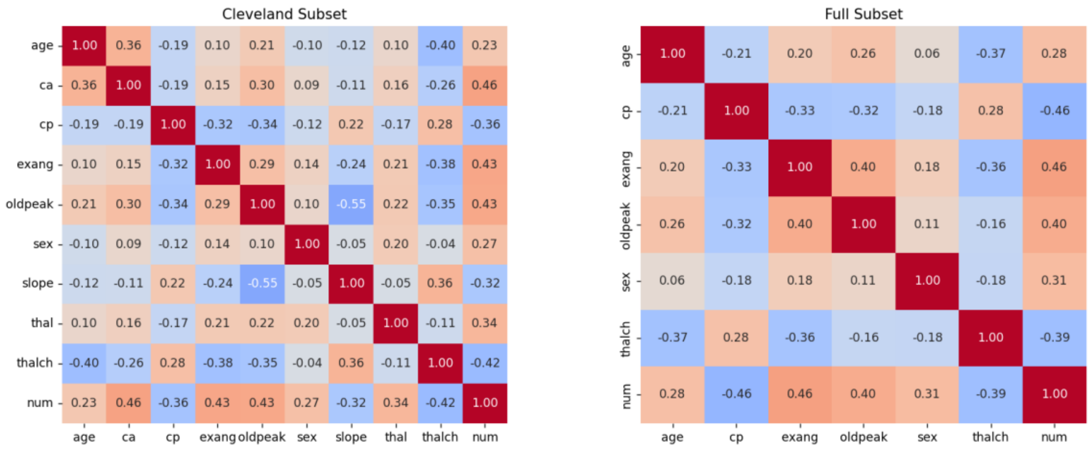

Figure 3. Correlation heatmap for both subsets, excluding features with absolute correlation below 0.2.

Figure 3 shows the correlation between features and the target variable *num*. The strongest associations are observed for *ca*, *exang*, *oldpeak*, and *thalch*, each with an absolute correlation of approximately 0.4. This pattern is consistent across both the Cleveland and full subsets, suggesting stable predictive relationships for these variables. However, lower correlation values do not imply that the remaining features are uninformative, as they may still contribute to predictive performance through non-linear effects or feature interactions.

2.2 Feature Description

Input Features:

1. age – numerical; age of patient in years
2. sex – categorical; male/female
3. cp – categorical; chest pain type (typical angina, atypical angina, non-anginal, asymptomatic)
4. trestbps – numerical; resting blood pressure in mm Hg on admission to the hospital
5. chol – numerical; serum cholesterol in mg/dl
6. fbs – boolean; fasting blood sugar. Originally categorical (True/ False).
7. restecg – categorical; resting electrocardiographic results (normal, stt abnormality, lv hypertrophy)
8. thalch – numerical; maximum heart rate achieved
9. exang – boolean; exercise-induced angina. Originally categorical (True/ False).
10. oldpeak – numerical; ST depression induced by exercise relative to rest
11. slope – categorical; the slope of the peak exercise ST segment (upsloping, flat, downsloping)
12. ca – numerical; number of major vessels (0-3) colored by fluoroscopy
13. thal – categorical; [normal; fixed defect; reversible defect]
14. num - the predicted attribute (0 = no heart disease; 1 = heart disease)

The features come directly from dataset, UCI Heart Disease Data [3]. No features were engineered. Full dataset subset removed ca, thal and slope features, while Cleveland kept them all.

3. Pipeline and Experiments

3.1 Preprocessing

The preprocessing pipeline is consistent across both dataset subsets and is implemented through two functions: *clean_data* and *preprocess_fold*.

The *clean_data* function is responsible for dataset-level cleaning and subset creation. It uses a Boolean parameter (*only_cleveland*) to either restrict the dataset to records originating from Cleveland or to retain all records while removing features with high levels of missing data. It also identifies and converts invalid physiological values (e.g., cholesterol or resting blood pressure equal to 0) into missing values (NaN). Additionally, binary categorical variables are encoded into 0/1 format to facilitate subsequent processing. These operations are applied prior to cross-validation, as they do not depend on fold-specific statistics and therefore do not introduce data leakage. This also improves computational efficiency by avoiding redundant processing within each fold. Features with more than 30% missing values (*slope*, *ca*, *thal*) were removed due to the risk of bias introduced through imputation. Although cholesterol (*chol*) also contains a moderate proportion of missing values (~22%), it was retained due to its clinical relevance in cardiovascular risk assessment. Despite its low linear correlation with the target variable ($|r| < 0.2$), cholesterol was not excluded, as correlation alone is insufficient for feature elimination in medical datasets where relationships may be non-linear or interaction-dependent. It is also noteworthy that missing cholesterol values occur only in the full dataset, which exhibits a higher overall level of missingness than the Cleveland subset.

The *preprocess_fold* function handles all transformations that require statistics derived from the data and therefore must be computed within each cross-validation fold to avoid data leakage. All parameters are learned from the training partition and then applied to both training and validation sets. The pipeline includes imputation of categorical variables using the most frequent class, one-hot encoding of nominal variables, median imputation for numerical features, Winsorization to reduce the impact of extreme outliers, and feature scaling to ensure comparable feature ranges. These steps ensure compatibility with the selected machine learning models and improve numerical stability during training.

3.2 Feature Selection and Engineering

Figure 4 demonstrates the permutation importance of the features. The results indicate that *ca* has substantially higher importance than the remaining variables, suggesting that the model relies heavily on the number of major vessels observed during diagnosis. Features such as *cp*, oldpeak, thal, fbs, exang and thalch also contribute to prediction performance, although to a lesser extent. No additional feature engineering was performed, as the original clinical variables were considered sufficiently informative. Following data cleaning and missing value handling, all remaining features were retained for model training.

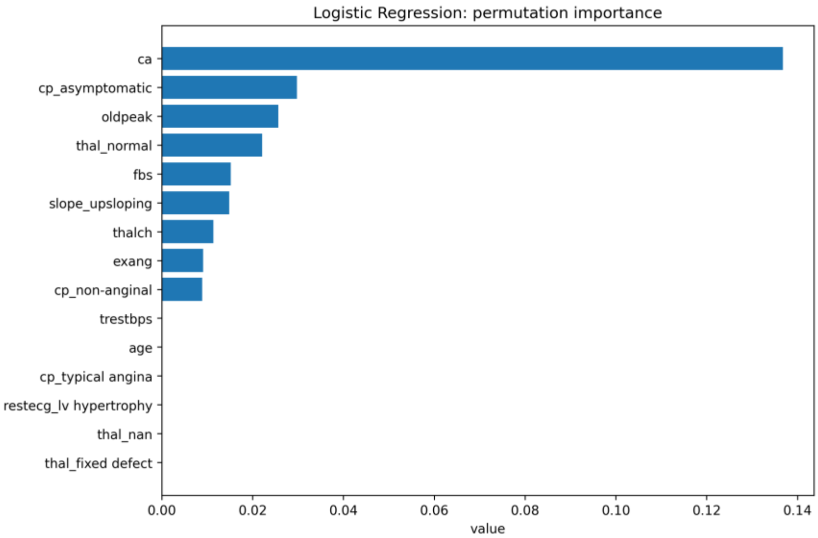

Figure 4. Permutation importance of predictors for the best-performing Logistic Regression model trained on the Cleveland dataset.

3.3 Model Selection and Cross-Validation

Table 3 shows the comparison of the candidate models. Three models were compared: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was used as a simple and interpretable baseline model. Decision Tree was added because it can model non-linear relations between features. Random Forest was used as a stronger ensemble model, because it combines many trees and is usually more stable than a single Decision Tree. Each model was tested on four experimental variants: Full unbalanced, Full balanced, Cleveland unbalanced, and Cleveland balanced. This allowed us to compare not only the models, but also the effect of dataset choice and class balancing.

Table 3. Comparison of candidate models.

Model	Hyperparameter tuning	CV Strategy	Val. Metric
Logistic Regression	Grid Search	5-fold CV stratified	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Decision Tree			
Random Forest			

3.4 Hyperparameter Optimization

Hyperparameter tuning was performed using Grid Search together with stratified 5-fold cross-validation. This means that several parameter values were tested, and each setting was evaluated across five folds. For Logistic Regression, parameter C was tuned. This parameter controls regularization: smaller C values apply stronger regularization, while larger C values allow the model to fit the data more closely. For each model, the parameter combination with the highest mean F1-score across five folds was selected. Smaller C values create a simpler model with stronger regularization, while larger C values allow the model to fit the data more closely. For Decision Tree, the maximum tree depth and minimum number of samples in a leaf were tuned. These parameters control how complex the tree can become. For Random Forest, the number of trees, maximum depth, number of features considered at each split, and minimum samples per leaf were tuned.

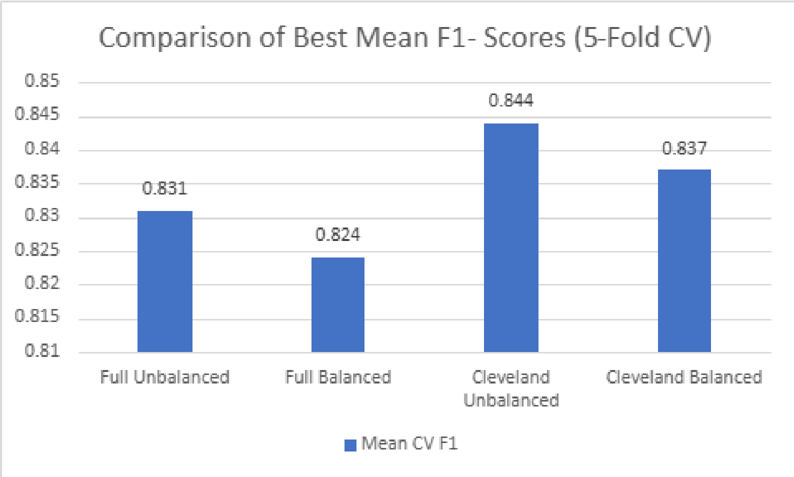

Figure 5. Best mean 5-fold cross-validated F1-score obtained by each model in each experimental configuration after hyperparameter optimization.

Figure 5 summarizes the highest mean cross-validated F1-score achieved by each model after hyperparameter optimization. The figure allows comparison of model performance across both dataset variants and balancing strategies.

3.5 Statistical Tests

Comparison	Mean F1	Wilcoxon p	t-test p	Interpretation
Full RF unbalanced vs balanced	0.831 vs 0.824	0.125	0.041	Small advantage for unbalanced
Cleveland LR unbalanced vs balanced	0.844 vs 0.837	0.438	0.387	No significant difference
Cleveland LR vs Full RF	0.844 vs 0.831	0.625	0.680	No significant difference

Table 4: Results of statistical comparisons based on fold-wise F1-scores.

Statistical tests were performed to determine whether the observed differences in model performance were statistically significant. The results of all statistical comparisons are summarized in Table 4. The primary comparison focused on the two best-performing models, Logistic Regression and Random Forest, evaluated on the Cleveland dataset without balancing. Logistic Regression achieved a mean F1-score of 0.844, while Random Forest achieved 0.830. A Wilcoxon signed-rank test and a paired Student's t-test were applied to the fold-wise F1-scores obtained during cross-validation. As shown in Table 4, the resulting p-values were 0.313 and 0.251, respectively. Since both values were greater than 0.05, the null hypothesis could not be rejected. Although Logistic Regression achieved a slightly higher average F1-score, the difference was not statistically significant. Additional statistical comparisons were performed to evaluate the effect of dataset balancing and dataset choice. The balanced and unbalanced variants were compared separately for the Full and Cleveland datasets using fold-wise F1-scores. As presented in Table 4, the Wilcoxon signed-rank test indicated no statistically significant differences between balanced and unbalanced variants in either dataset ($p > 0.05$). Similarly, the comparison between the best-performing Cleveland and Full configurations showed no statistically significant difference ($p > 0.05$).

3.6 Final Model Training

Due to the limited size of the dataset, model evaluation was performed using Stratified 5-Fold Cross-Validation. Cross-validation allows all observations to contribute to both training and validation while preserving class proportions within each fold. Since the objective of this study was to compare the relative performance of different models and preprocessing strategies rather than estimate deployment performance on a completely unseen population, a separate hold-out test set was not used. The reported results are the means from final k-fold validation with tuned parameters.

4. Results and Discussion

4.1 Performance Evaluation

Table 5: Results Summary

Dataset	Balancing	Best model	Accuracy	Precision	Recall	F1	ROC-AUC
Cleveland	unbalanced	Logistic regression	0.862	0.874	0.820	0.844	0.917
Cleveland	balanced	Logistic regression	0.852	0.849	0.827	0.837	0.914
Full	unbalanced	Random forest	0.805	0.801	0.864	0.831	0.874
Full	balanced	Random forest	0.798	0.796	0.855	0.824	0.870

Table 4 summarizes the best-performing model for each experimental configuration. The results show three main findings. First, the Cleveland dataset performed best overall, even though it contained fewer samples. Second, class balancing did not improve the results. Third, Logistic Regression and Random Forest achieved the best performance, while Decision Tree was generally the weakest model. The best overall result was obtained by Logistic Regression on the unbalanced Cleveland dataset, with an F1-score of 0.844 and ROC-AUC of 0.917. The best Full dataset result was obtained by Random Forest, with an F1-score of 0.831 and ROC-AUC of 0.874. The Cleveland model achieved an F1-score higher by 0.013 and a ROC-AUC higher by 0.043 than the best Full dataset model. A possible explanation is that the Cleveland dataset retained important clinical features such as ca, thal, and slope, which had to be removed from the Full dataset because of missing values.

Table 6: Influence of class balancing on model performance - summary

Dataset	Metric	Unbalanced	Balanced	Difference
Full	F1	0.831	0.824	-0.007
Full	ROC-AUC	0.874	0.870	-0.004
Cleveland	F1	0.844	0.837	-0.007
Cleveland	ROC-AUC	0.917	0.914	-0.003

Table 5 shows that class balancing produced only small changes and did not improve performance. In both datasets, the unbalanced versions achieved slightly higher F1-score and ROC-AUC. Since the original class distributions were already quite similar, additional balancing was not necessary and slightly reduced predictive performance.

Figure 6 presents the confusion matrix and ROC curve for the best-performing model, Logistic Regression trained on the unbalanced Cleveland dataset. The confusion matrix shows that the model correctly classified the majority of both healthy and diseased patients, while the ROC curve demonstrates strong discrimination ability. The ROC-AUC value of 0.917 indicates that the model was able to distinguish effectively between the two classes. These results are consistent with the high F1-score reported in Table 4.

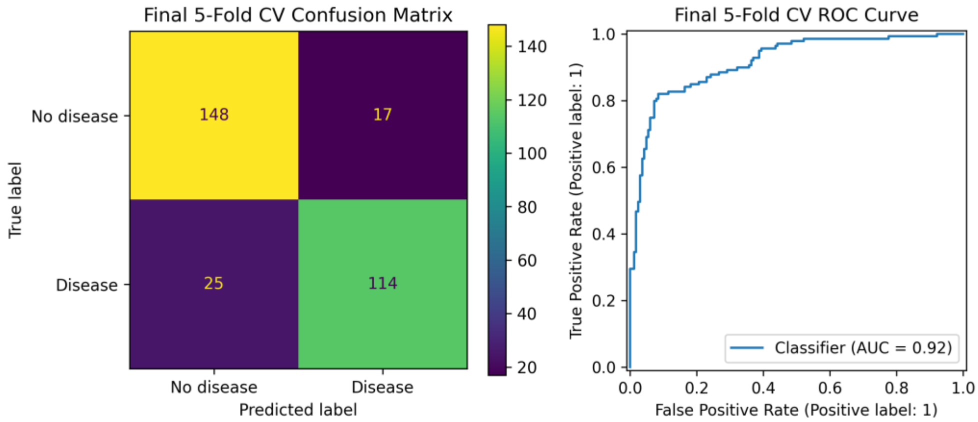

Figure 6. Confusion matrix and ROC curve obtained from the final 5-fold cross-validation evaluation of Logistic Regression on the unbalanced Cleveland dataset.

4.2 Model Explainability

To better understand the model predictions, feature importance was analysed for Logistic Regression and Random Forest. For Logistic Regression, the most important variables were ca, cp_asymptomatic, exang, and oldpeak. For Random Forest, important features included thalch, age, cholesterol, oldpeak, and asymptomatic chest pain.

Although the exact rankings differed between methods, the same clinically meaningful variables appeared repeatedly. Features related to chest pain, exercise response, heart function, and vessel abnormalities had the greatest influence on predictions. This suggests that the models learned medically meaningful patterns rather than random relationships. Figures 7 and 8 show the results for the logistic regression model.

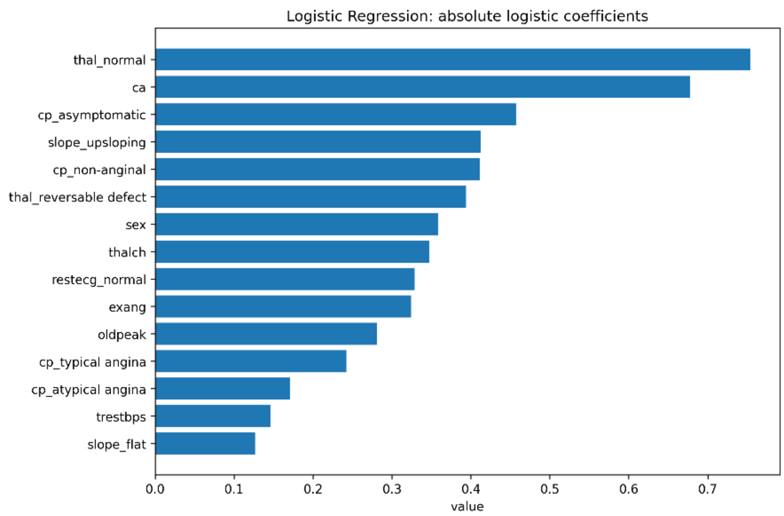

Figure 7. Feature importance based on the absolute Logistic Regression coefficients for the unbalanced Cleveland dataset.

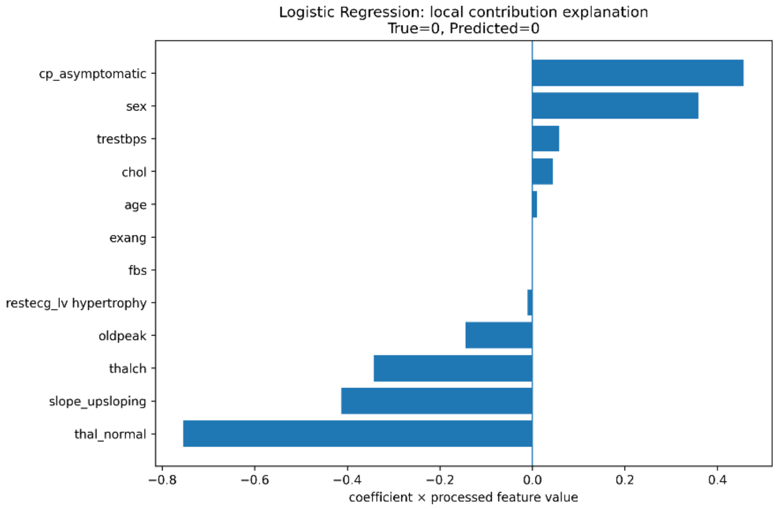

Figure 8. Local explanation for an individual prediction generated by Logistic Regression on the unbalanced Cleveland dataset.

4.3 Error Analysis

Despite the strong overall performance, classification errors were observed across all evaluated models. False positives may result in unnecessary follow-up examinations, whereas false negatives are more critical because they may delay diagnosis and treatment. Some errors are likely because of overlapping characteristics between healthy and diseased patients, making classification inherently difficult. Additionally, the limited size of the dataset may restrict the amount of information available for learning predictive patterns. In medical diagnosis, recall is prioritized to reduce false negatives.

5. Conclusions

The best performance was achieved by Logistic Regression trained on the unbalanced Cleveland dataset. This shows that a simple, interpretable model can achieve strong predictive performance for this task. A key finding was that retaining clinically relevant features had a greater impact on performance than increasing dataset size, as the smaller Cleveland dataset consistently outperformed the larger full dataset. This is important as it shows that a more complex model is not always better. When the dataset contains strong clinical features, a simpler model can perform very well.

It is important to note, however, that the statistical tests showed that the differences between datasets and balancing approaches were not significant. This shows that all tested approaches can yield acceptable results, especially with careful engineering. Class balancing had minimal effect on results, possibly due to small difference of size between classes. Feature importance analysis identified clinically meaningful predictors such as the number of major vessels observed by fluoroscopy (*ca*), chest pain type, exercise-induced angina, and ST depression, aligning with established cardiovascular risk factors. Future research should include testing on different datasets, different models and evaluation in clinical settings. Overall, the results demonstrate that machine learning can effectively support heart disease prediction using routine clinical variables and simple models.